

Technical Document #210: Cybersecurity - Comprehensive Overview

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Network segmentation divides networks into smaller segments. It limits the spread of attacks and improves security monitoring.

Incident response plans define how organizations respond to security incidents. They include preparation, detection, containment, and recovery phases.

Zero trust architecture assumes no implicit trust and verifies every request. It minimizes the attack surface and limits lateral movement.

Penetration testing simulates cyber attacks to identify vulnerabilities. It helps organizations assess their security posture before real attacks occur.

Multi-factor authentication (MFA) requires multiple forms of verification. It significantly reduces the risk of unauthorized access.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Key Takeaways:

- Intrusion detection systems (IDS) monitor network traffic for suspicious activity.
- Multi-factor authentication (MFA) requires multiple forms of verification.
- Zero trust architecture assumes no implicit trust and verifies every request.